Muhammad Danish

PhD Student, Computer Science, *University of New Mexico*
Albuquerque, NM • mdanish@unm.edu • +1 (505) 219-6410 • mdanish.net

Research overview

I study measurement-driven security, privacy, and fraud at the Internet scale. I build end-to-end pipelines that combine automated data collection pipelines, LLM-based security decisions, and manual human analysis. In the near term, I aim to work on AI/ML agents for large-scale detection and response across security, privacy, and fraud.

Education

- **Ph.D. Computer Science**
  University of New Mexico, Albuquerque, NM, USA — Aug 2025 – May 2028
- **M.S. Computer Science**
  University of New Mexico, Albuquerque, NM, USA — Aug 2025 – May 2026
- **B.S. Computer Science, Honors Interdisciplinary Liberal Arts Minor**
  University of New Mexico, Albuquerque, NM, USA — Aug 2021 – May 2025

Publications

1. **Danish, M.**, Sobrados, E., Kaushik, P., Acharya, B., Saad, M., Mueen, A., Rahaman, S., & Anwar, A. "**Private Links, Public Leaks: Consequences of Frictionless User Experience on the Security and Privacy Posture of SMS-Delivered URLs**". *(Under Submission - IEEE S&P 2026)*
   322K URLs analyzed from 33M SMS messages. Built an end-to-end pipeline (data collection, LLM-based PII detection, and manual validation) to identify 150+ PII-exposing services. Uncovered weak bearer-token authentication, low-entropy token user enumeration, privileged-action exposures, account takeover, and additional PII via APIs/WebSockets/HTML bootstrapping. Responsible disclosures resulted in fixes protecting 150M+ users.
2. Acharya, B., **Danish, M.**, Pape, D., Saad, M., Anwar, A., Schönherr, L., & Holz, T. "**Surveyception: An Exploration of Deceptive Survey Forms**". *(Under Submission - ACM IMC 2026)*
   140K survey forms collected from 10 major providers for large-scale detection of sensitive-data collection and deceptive survey behavior via LLM-assisted classification, fraud-signal correlation, and expert review. Identified 2,645 sensitive-info forms, including 566 scamming forms. Performed scammer network analysis, uncovering 3,200+ accounts participating in 300+ coordinated scam campaigns.
3. "**Fraudfunding: Measuring Fake Charity Donations in Crowdfunding Platforms**". *(In Progress)*
   Cross-platform measurement across 30+ fundraising sites; multi-step analysis of campaign data spanning disaster relief, health, and animal causes; responsible disclosure to platforms and PayPal.
4. "**A Large-Scale Study of Legacy VPN Protocol Use and Misconfigurations**". *(In Progress)*
   67K+ enumerated VPN servers across 15+ mobile apps; reversed 20+ private API endpoints; measurement of legacy protocol usage and weak configurations; responsible disclosure to affected vendors.
5. "**GeoVPN: Country-wise Analysis of Mobile VPN Apps**". *(In Progress)*
   Comparative analysis of mobile VPN apps across countries; collect and analyze data at scale; downstream analysis of cross-country differences; early takeaways informing user risk and policy guidance.

Experience

- **Research Assistant (PI and Reference: Afsah Anwar), University of New Mexico** — Dec 2023 – Present
  - Conducted security research on scams and fraud using large datasets, resulting in multiple publications.
  - Designed multi-layered security checks to detect vulnerabilities in online ecosystems.
  - Enhanced project outcomes and knowledge sharing via research collaborations with academia and industry.

- **Student Programming Analyst, University of New Mexico** Oct 2022 – Aug 2025
  - Achieved ∼200% efficiency in campus processes using Power Automate, Power Platform, and Boomi.
  - Increased project visibility and online presence by developing websites using Cascade CMS and WordPress.
  - Reduced testing times by ∼50% through Playwright and Azure DevSecOps pipelines.
  - Managed cross-functional project implementations across multiple departments for 20,000+ students.
- **Project Manager, KritiKal Solutions (Remote)** May 2021 – Aug 2022
  - Led cross-functional engineering teams, streamlining operations and driving successful project execution.
  - Reduced project duration by ∼20% by efficiently managing requirements, timelines, and progress tracking.
  - Saved ∼$1,000/month in subscriptions and optimized workflows by developing an internal knowledge base.
  - Enhanced decision-making by building a real-time data visualization app, providing actionable insights.

Teaching

- **Computer Science Teaching Assistant, University of New Mexico**
  - Enhanced students' understanding by preparing assignments and exams and providing valuable feedback.
  - Raised class performance through hands-on coding workshops, engaging over 40 students in weekly labs.
  - Fostered a collaborative learning environment by mentoring students via group discussions, one-on-one sessions, and YouTube.

- **Graduate Teaching Assistant, University of New Mexico**
  CS544: Introduction to Cybersecurity — Spring 2026
- **Graduate Teaching Assistant, University of New Mexico**
  CS585: Computer Networks — Fall 2025
- **Guest Speaker, University of New Mexico**
  CS544: Introduction to Cybersecurity — Spring 2025
- **Undergrad Teaching Assistant, University of New Mexico**
  CS251L: Intermediate Programming (Java) — Fall 2023
- **Undergrad Teaching Assistant, University of New Mexico**
  CS251L: Intermediate Programming (Java) — Spring 2023
- **Undergrad Teaching Assistant, University of New Mexico**
  CS105L: Introduction to Computer Programming (Python) — Fall 2022

Awards & Recognitions

- **CRA Outstanding Undergraduate Award** - North America
- **New Mexico Engineering Foundation Scholar** - New Mexico
- **Tallant Cooper Scholar** - University of New Mexico

Skills

- **Programming:** Bash/PowerShell, Python, PyTorch, Java, JavaScript, HTML, CSS, SQLite, MATLAB
- **Tools/Platforms:** UNIX/Linux, Playwright/Selenium, LLMs, NMap, Wireshark, Burp Suite, Kibana, Metasploit, Azure, Git, Cascade, WordPress, Cherwell
- **Technical:** Large-scale data analytics, web security, AI/ML agents, LLM prompt engineering, data scraping

Volunteering

- **Section Leader (Stanford Code in Place):** Mentored diverse students in Python and CS fundamentals.
- **Artifact Evaluation Committee (IEEE S&P, PETS and ACSAC):** Reviewed top-tier research artifacts
- **LoboSec (VP):** Organized weekly trainings and campus outreach to compete in collegiate and online CTFs.
- **CSGSA (Treasurer):** Managed budgets and reimbursements to fund workshops and events.